

## Wk 3 Lab log

### Wk 3 Summary

This week involved general debugging and a lot of reading trying to figure out how to extract critical exponents from data.

### Lab log 05/02/2026:

Quick group meeting w/Dr Turci; outlined report requirements, structure and expectations

Discussed w/Scarlett direction; she's more interested in cluster behaviour, i'm more interested in critical exponents. Most of the core code functionality is setup; so there may be a point at which we start to diverge in our research interests. TBD, more code experimentation and setup to do, but to be considered.

quickly attempted to generate data over temperature; however `1/np.array` doesn't create a list of reciprocals; so that was a mistake

am generally fiddling with data ranges and plotting graphs to get an idea of what data ranges would be ideal for this project. generally speaking range  $\beta \in [0, 3]$  seems appropriate.

Have implemented temperature normalisation (see "Wk 3 Additional Reading"); the absolute magnetisation over beta plots change from:

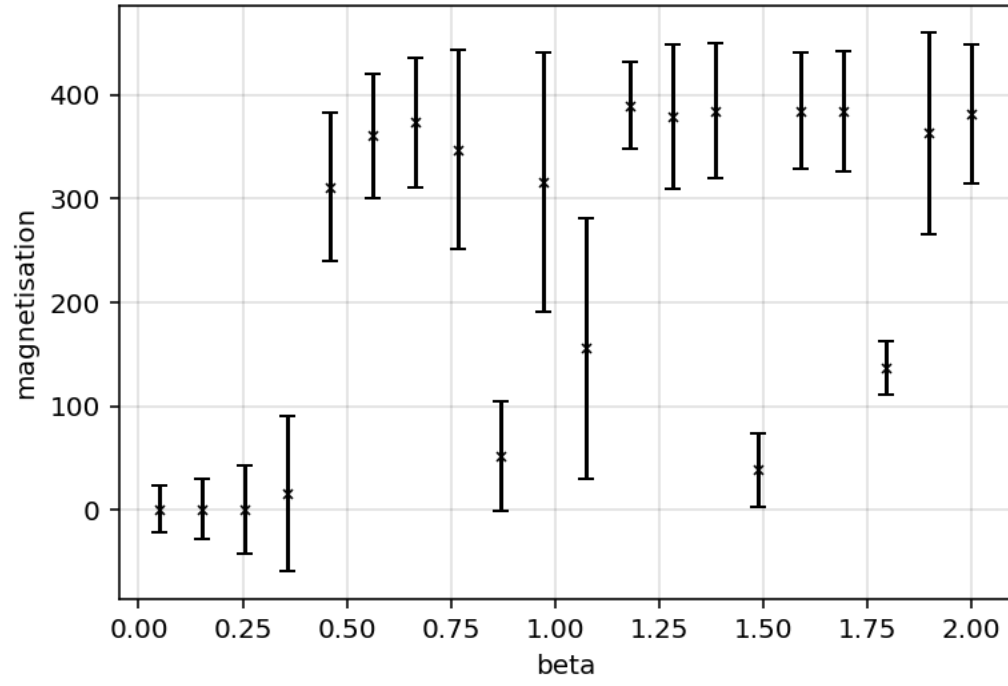


Figure 1: Fig 3.1 -  $M(\beta)$  before implementing Eq. 3.1 for  $(d = 2, L = 20)$

to

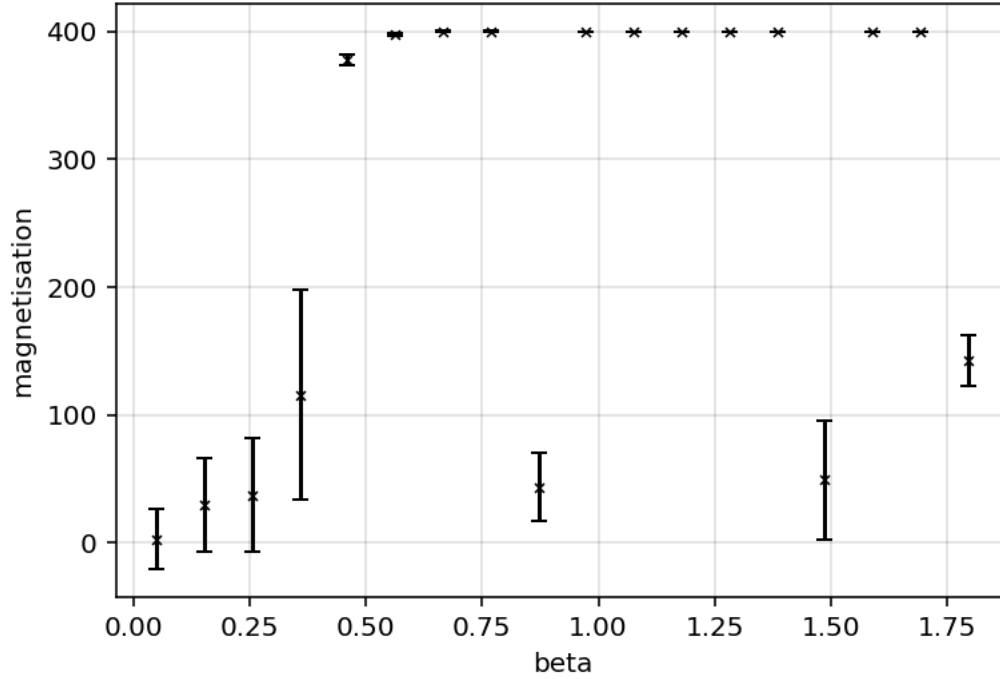


Figure 2: Fig 3.2 -  $M(\beta)$  after implementing Eq. 3.1 for  $(d = 2, L = 20)$

This change seems to better incorporate the probability of configs, thus “converging” the average magnetisation to the “true” magnetisation significantly faster and better. This has made the phase transition a lot clearer; in  $(d = 2, L = 20)$  appears to be  $\sim \beta_c = 0.4$  - note however that equilibration times are a lot longer near criticality, so is noisier

#### To do list:

- implement variable equilibration steps into dataset (so can pass in either float, or array of equilibration values) (done)
- normalise observables
- Figure out way to compute critical  $\beta_c$  from data
- do a basic fit to  $|T - T_c|^x$  where  $x$  is the critical exponent
  - examine basic  $x$  over a range of sizes to establish finite size effects
  - try and figure out way to account for finite size effects
- repeat analysis in higher dimensions

### lab log continued

have tested a narrower temperature range ( $T \in [0.5, 5]$ ), yields:

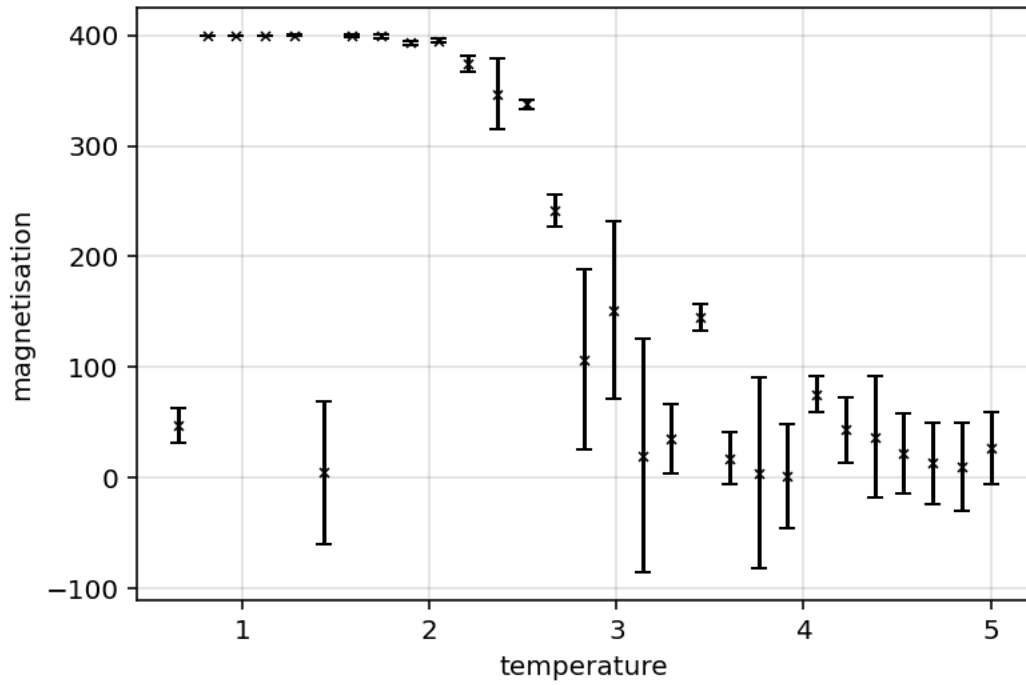


Figure 3: Fig 3.3 - absolute magnetisation over temperature for ( $d = 2$ ,  $L = 20$ )

Note how high temperatures seemingly have a lot more uncertainty.

Also generated plots for the other observables:

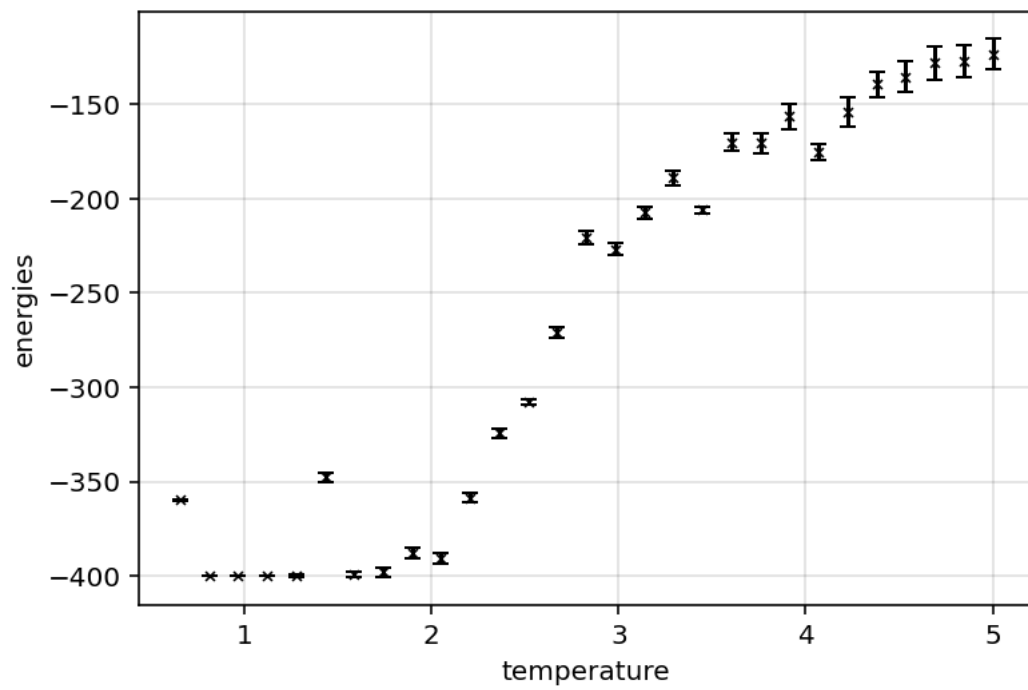


Figure 4: Fig 3.4 - energies ( $d = 2$ ,  $L = 20$ )

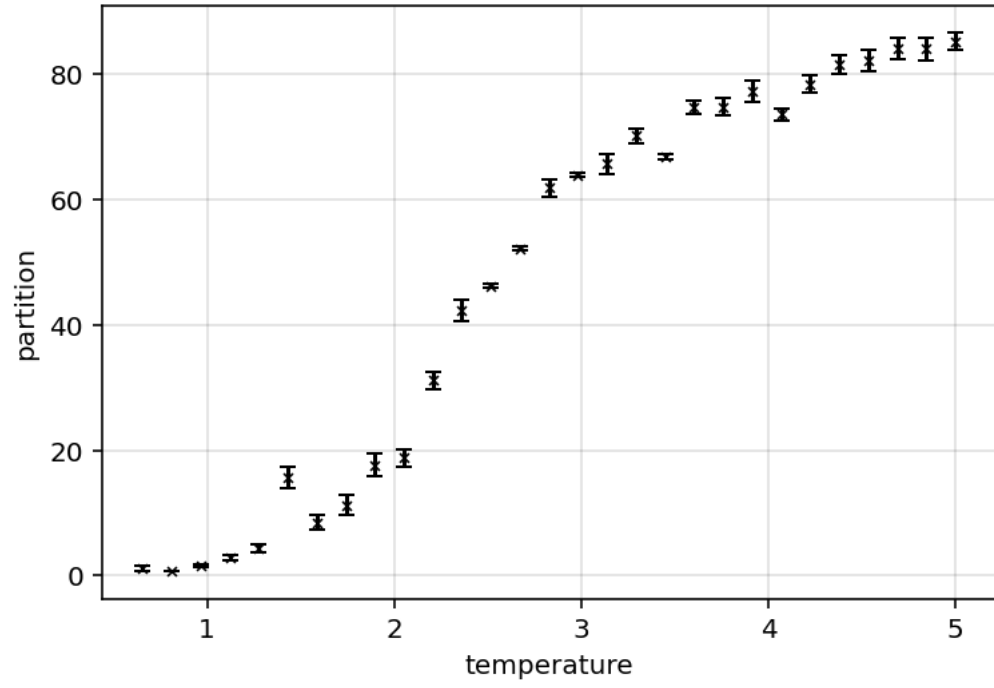


Figure 5: Fig 3.5 - partition function ( $d = 2$ ,  $L = 20$ )

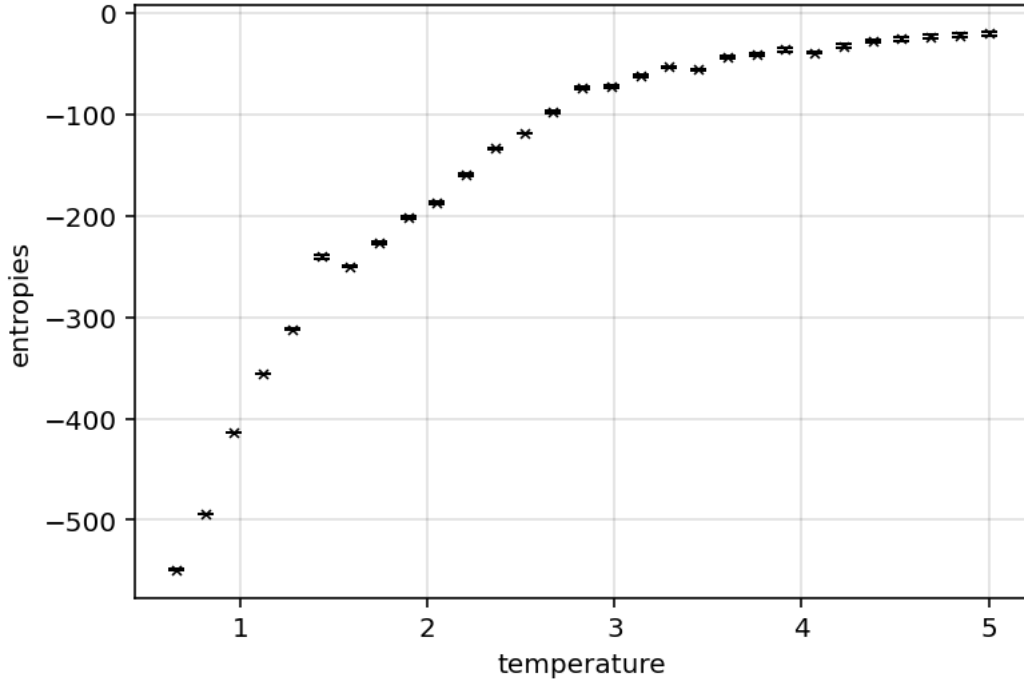


Figure 6: Fig 3.6 - entropies ( $d = 2$ ,  $L = 20$ )

Note how Fig 3.4 seems to have some asymmetry in it's transition. It appears to have a near discontinuous increase about  $T \approx 2$ , but has a smoother continuous change as  $T$  increases. A few potential explanations worth considering:

- From (Christensen and Moloney 2005), it seems that the  $|T - T_c|^x$  expressions are usually one sided; this may be a manifestation of that?
- May be finite size effect

hemlholtz plot finally produced (code errors required re-simulating data for useful results):

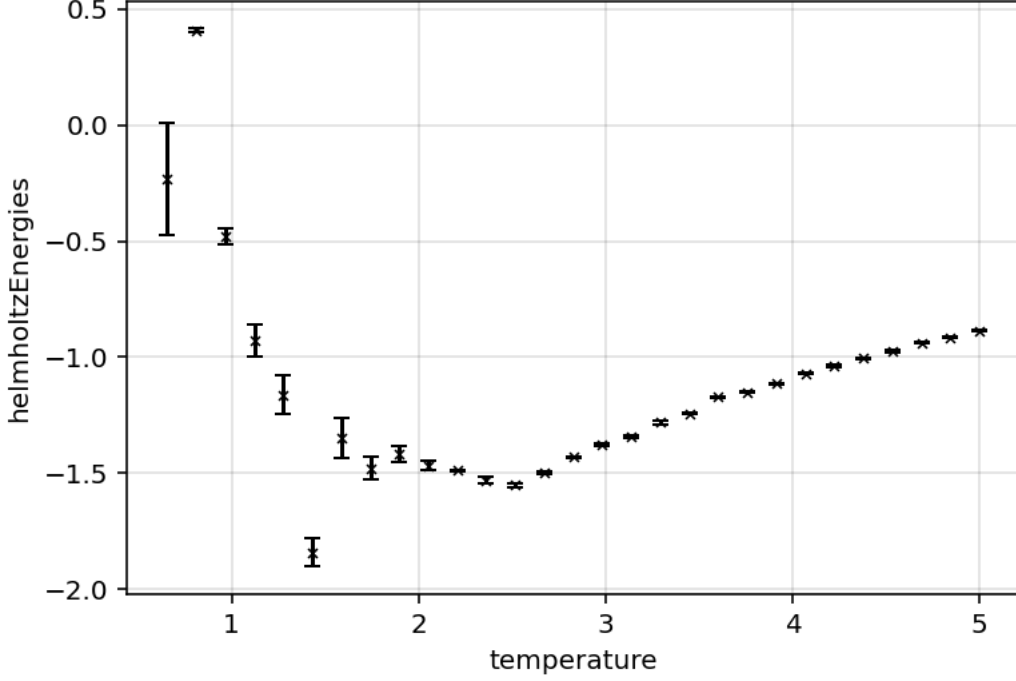


Figure 7: Fig 3.7 - Helmholtz free energies ( $d = 2$ ,  $L = 20$ )

It seems that the critical temperature coincides w/minima of helmholtz free energy (ignoring outlying data points). Unsure if this is coincidental or not, however if so, may provide an incredibly easy method of finding the critical temperature.

#### dataset data analysis:

Starting to think about how to analyse the data being generated. Particularly, need to do following:

- find the critical temperature
- fit exponent curves of form  $|T - T_c|^x$  to observables as appropriate (they may be one sided)

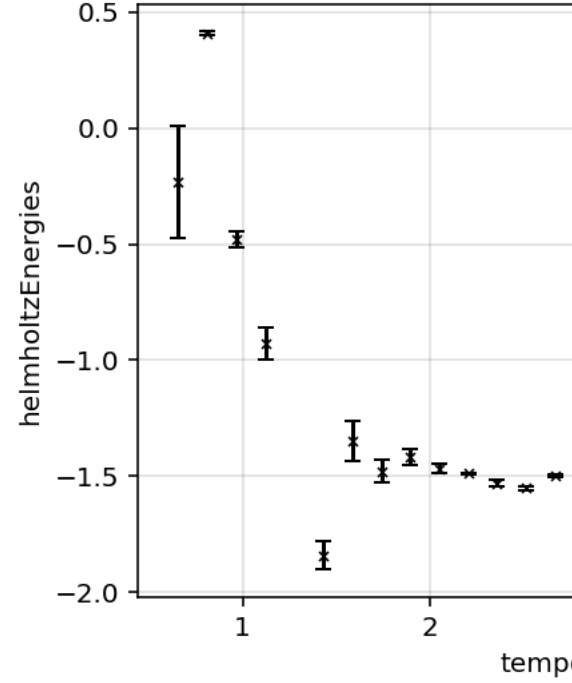
To do so however it is necessary to filter for evident outliers in the data. This may be done manually, however it'd be more convenient to do so automatically. A potential means to do this would be to filter by a generic “continuity check”. Given a particular observable  $a_i$  with neighbours  $a_{i-1}$ ,  $a_{i+1}$ ; it would interpolate between the neighbours; if  $a_i$  is within uncertainty of the interpolated value, it is accepted, else it is rejected. This approach may cause erroneous results however when evaluating the edges of the dataset. Since this method relies on neighbours,



it can't validate data at either end of the dataset. It may be possible to extrapolate to the data edges from the next 2 nearest points?

That may be a better way of doing it; extrapolate from the 2 nearest points for the continuity check. For most datapoints that will be both neighbours, however for the endpoints it will extrapolate from the data.

This method may be a bit crude, but should discard most outliers. The main risk is that it may “flatten” trends in the data due to datapoints being too far apart from each other. However adequate sampling density should avoid this issue.



Quickly implementing this method on the helmholtz energies yields:

Which seems to not work. minor fixes; code is UGLY, but it seems to work. Filtering seems to work.

Talked to Dr Turci, asked about the apparent minima in helmholtz free energy; pointed out its a non standard way of finding critical temps; suscpetibility/spin and specific capacity/spin are more common:

$$\chi = \beta \frac{\langle M \rangle^2 - \langle M^2 \rangle}{N} \quad \text{Eq. 3.2}$$

$$C = \beta^2 \cdot k_b \cdot \frac{\langle E \rangle^2 - \langle E^2 \rangle}{N} \quad \text{Eq. 3.3}$$

attempted implementation; doesn't seem to be working yet. needing to resimulate the data. Am also unsure how to propagate errors through the susceptibility and heat capacity either. propagating errors in terms yields:

$$\Delta\chi^2 = (\frac{\beta}{N})^2((2\langle M \rangle)^2\Delta\langle M \rangle^2 + \Delta\langle M^2 \rangle^2) \quad \text{Eq. 3.4}$$

$$\Delta C^2 = (\frac{\beta^2}{N})^2((2\langle E \rangle)^2\Delta\langle E \rangle^2 + \Delta\langle E^2 \rangle^2) \quad \text{Eq. 3.5}$$

susceptibility seems to be working-ish (errors are massive but given expression i'm not sure error propagation makes sense as a concept) - also peak of susceptibility seems to be offset relative to apparent critical temperature in all other graphs ( $T_{Cs} \approx 3$ ).

Error found. Eqs 3.2-3.5 use the mean, code was using weighted average instead. Quick graphs produced:

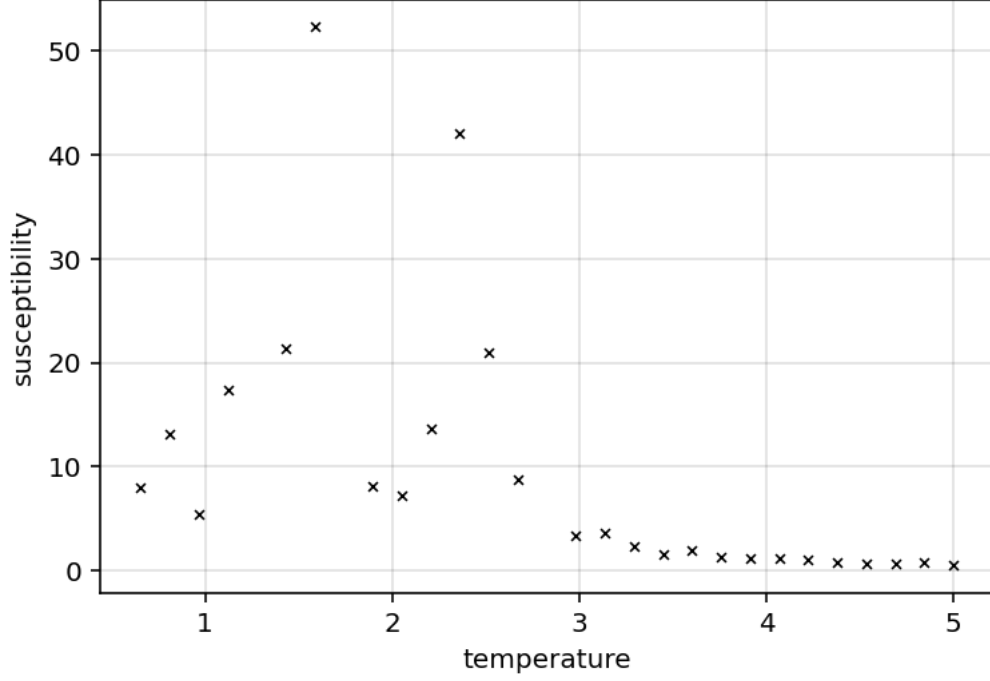


Figure 8: Fig 3.8 - Susceptibility ( $d = 2, L = 20$ )

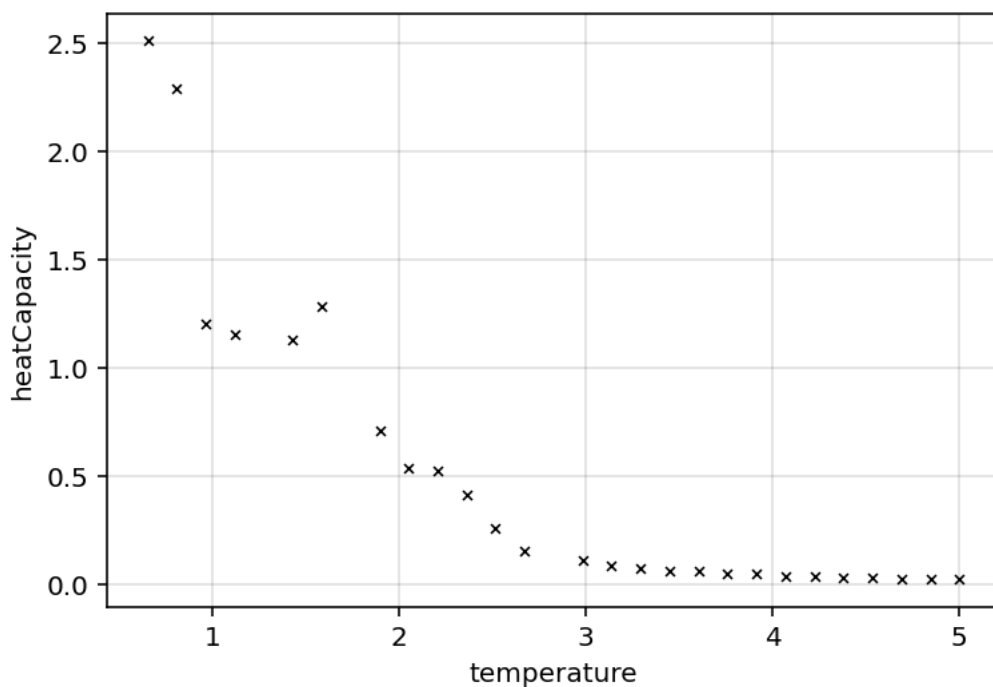


Figure 9: Fig 3.9 - Heat capacity ( $d = 2$ ,  $L = 20$ )

Don't think these work fully. actually unsure if i should used weighted average; textbook seems to indicate weighted average? All of these tests have been conducted on a single dataset (saved for convenience) - it may be worth testing on a larger, more fine grain dataset.

#### To Do list:

- fix susceptibility and heat capacity computations
- run big dataset; see if that fixes
- figure out exponent fitting
- add a “recompute observables” method to config object; that way old config objects can be updated if computations change
- double check all observable computations

#### Lab log 06/02/2026:

Not doing proper lab work today, but am generating some larger datasets in the background (40 configs over  $T \in [0.5, 5]$  for 100x100, 1000 Equil steps, 500 sim steps) to have better testing/development data. Furthermore, Dr Turci did some more thinking on the apparent

minimal helmholtz free energy cooresponding to critical temperature. Apparently the critical temperture should show up in the second derivative  $\frac{d^2 F}{dT^2}$ , but shouldn't be visible as is (given  $dF = pdV - SdT$ , and entropy SHOULD be positive (may need to double check thats computed correctly), F should decrease over data range); Dr Turci suggests it may be poor sampling about  $T_c$ ; i suspect i've computed an observable incorrectly.

## Lab log 07/02/2026:

observables implemented:

- Energy (code verified)
- Magnetisation (code verified)
- Partition function (code verified)
- Entropy (code verified)
- Helmholtz free energy (code verified)
- Boltzman weights (Given average observables are weighted by boltzman distribution, computing these weights seperately made sense) (code verified)
- Susceptibility (code verified)
- Heat capacity (code verified)

quickly rewrote calcEnergy - needed to be generalised; plus was opportunity to double check correct implementation Similarly rewrote partition function method as it used a very similar implementation to calcEnergy

Have double checked all the computation; nothing is immediately wrong reran data; seems susceptibility fixed itself, however heat capacity hasn't

I suspect it may be due to noise in the data? would need to run a larger dataset to verify.

Implementing "continuous saving" to "fromBetas" dataset construction - ensures that data saves over dataset generation ensuring interrupted runs don't wipe out whole chunk of data (also implemented "already exists" checks as part of this, allows for quick restarts) - code is messy but works and shouldn't need to be touched significantly

cleaned up code a bit

have started working in critical exponent fitting.

comparing values to <https://www.phys.ens.psl.eu/~jacobsen/AIMES/Ising.pdf> which quotes  $\beta = \frac{1}{8}$ ; testing finds  $\beta = 0.03$  for 20x20 config; off by a factor of 4. Not the worst result however, but the fit leaves much to be desired:

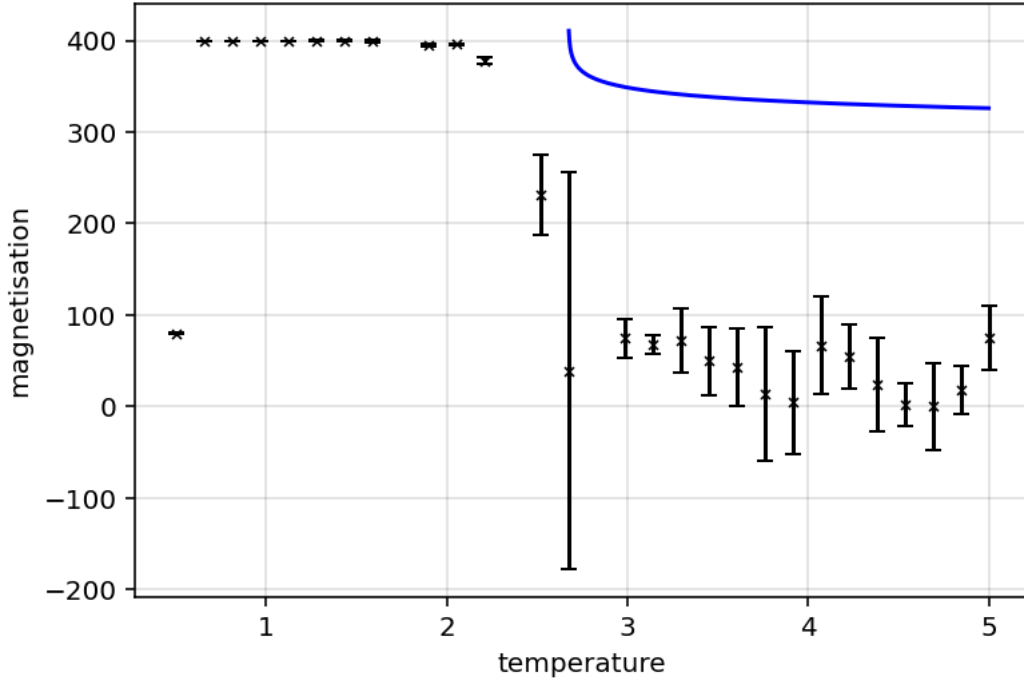


Figure 10: Fig 3.10 - Fit attempt to  $M \propto |T - T_c|^\beta$   $\beta = -0.03 \pm 0.05$  ( $d = 2$ ,  $L = 20$ )

Will need to do some debugging to see where this may be coming from...

have created new .py file; simply a place to streamline “bulk data generation” - imports everything else, that way desired datasets can be neatly laid out as a queue Furthermore, have updated how datasets save configs to files; they will also check if duplicate configs already exist and skip them if they do. This makes restarting interrupted datasets easier, and means that its easier to add data to an existing dataset (I.E, it may be useful to add a high density region around critical points with higher equilibration steps).

Am setting my laptop to generate some main data runs overnight; these are all 2D models in 3 batches which represent 1000, 2000, 3000 equilibration steps. The first 2 batchkes (1000, 2000 steps) have a range of sizes [20, 30, 40] whilst the final batch has a range [20, 30, 40, 50, 60, 70, 80, 90]. This was done to compare the effects of equilibration times, as well as to begin investigating finite size effects empirically. The longest equilibration dataset has more sizes because it is expected to be the highest quality data, and establishing the effect of equilibration can be done with a more limited subset of data.

## Lab log 09/02/2026

Laptop is still generating data. First night, it shut itself down interrupting progress. Second night it was kept on, and mostly handled the  $100 \times 100$  over 3000eq steps, 500sim steps. Starting at  $\sim 14:00$  08/02/2026, till now (10:00 09/02/2026) it has generated 18/40 datapoints. Given that each simulation calls “mcMove” for each step taken, and that “mcMove” loops through and performs a computation for each of its cells; the above timings gives a rough computation speed of  $8750 \text{ cellcomputations/s}$  (alternatively  $31.5 \cdot 10^6 \text{ cellcomputations/Hr}$  or  $0.756 \cdot 10^9 \text{ cellcomputations/day}$ ). This is useful as it allows for time estimates for future datasets to be planned in advance, allowing for better planning. The theoretical timings for potential datasets of 3000eq, 500sim steps, 40 datapoints follow:

- 3D  $20^3$ :  $\sim 1.48$  days
- 3D  $22^3$ :  $\sim 1.97$  days
- 4D  $10^4$ :  $\sim 1.85$  days
- 5D  $6^5$ :  $\sim 1.44$  days
- 6D  $4^6$ :  $\sim 0.76$  days
- 4D  $20^4$ :  $\sim 29.6$  days

The large 4D example is impractical to generate given the time constraints of this project. However all the other options aren't impractical. Combined they'd take  $\sim 1\text{Wk}$  non stop computation to complete. Whilst there are likely to be some interruptions, i think this may be feasible. The main question is to what degree can finite size effects be accounted for in such small model sizes; if its possible to account and “correct” for such finite size effects, comparing critical exponents over  $d \in [2, 6]$  may be interesting. However given the significant computational investment, its worth **knowing** the following:

- How to correct/account for finite size effects - this is crucial, otherwise the different sized different dimensioned datasets can't be compared to each other
- What should the dataset parameters be for arbitrary dimension/size:
  - How does model size and dimension relate to required equilibration steps
  - How does critical temperature relate to dimensionality
  - this is important because of the heavy computational investment; exploring a particular dimension to determine these parameters experimentally would be a waste of time; so trying to estimate the appropriate equilibration steps and temperature ranges.

As such, its worth doing some additional reading/research particularly to address correcting for finite size effects. This reading is notated in “Wk 3 Additional Reading”.

### plans from this point forwards:

As it stands, I think i have a better idea of the renormalisation theory, however an still somewhat unsure how to use it. Particularly i'm unsure how to find the  $y_t$ ,  $y_h$  scaling factors (i'm assuming that its scaling factors relative to a "baseline" size? so  $y_{\{0\}}$  would be value for baseline size, muliplied by  $y_{\{t\}}$  for any other size?).

As such I have 2 options to deal with critical exponents:

- Fit to  $t = \frac{T-T_c}{T_c}$  instead of  $T$  - this may already solve my issues in fig. 3.10
- Ask Dr Turci how to relate current understanding of renormalisation theory to the data i have

There are 2 possible outcomes of that:

- I can effectively renormalise the results of any sized configuration to an equivalent "standard size"
- I can't do that

If its possible to renormalise the results, it becomes possible to measure critical exponents and compare them between differently sized models. If this is the case, it may be possible to examine the exponents of significantly higher dimensions whilst simulating relatively small sizes. If so, it'll be necessary to setup a queue of desired higher dimensioned datasets as soon as possible, and run it near continuously. If not i may need to restrict my investigations to  $d = 2, 3$  for size  $L = 20$ .

At any rate, the next coding objective is to figure out the exponent expression fitting and create a robust method for measuring the critical exponents for all observables (heat capacity, magnetisation, susceptibility).

I'll also need to find further references for:

- values for critical exponents (wikipedia has an uncited table which is annoying)
- mean field theory derivations

Christensen, Kim, and Nicholas R Moloney. 2005. *Complexity and Criticality*. Imperial College Press. <https://doi.org/10.1142/p365>.